

Unsupervised ensemble pruning technique for Isolation Forest

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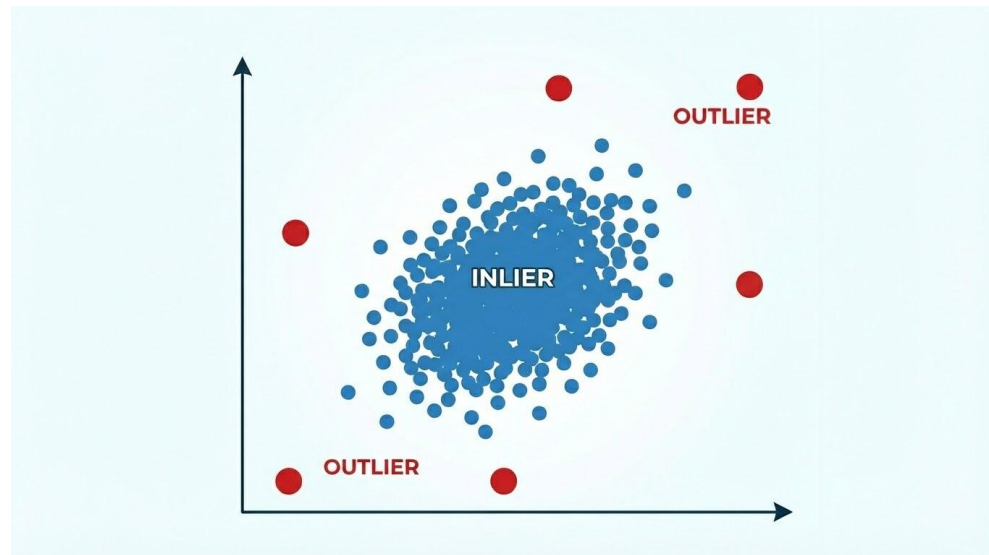


Background - Anomaly Detection



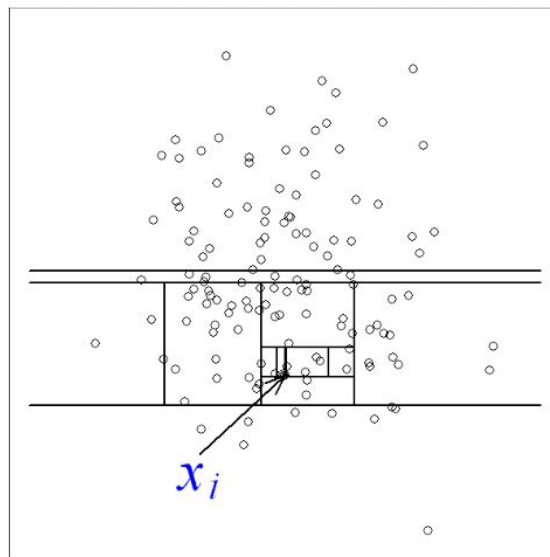
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- Problem of discriminating **normal** and **anomalous** data
- Real-world applications
 - healthcare
 - finance
 - cybersecurity
 - etc...

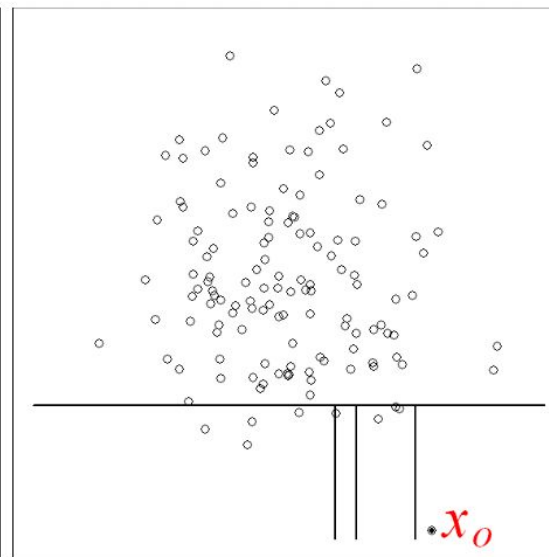


Background - Isolation Forest

- Foundational concepts:
 - Outliers are **rare events**
 - Outliers have significant **different attribute values** w.r.t inliers



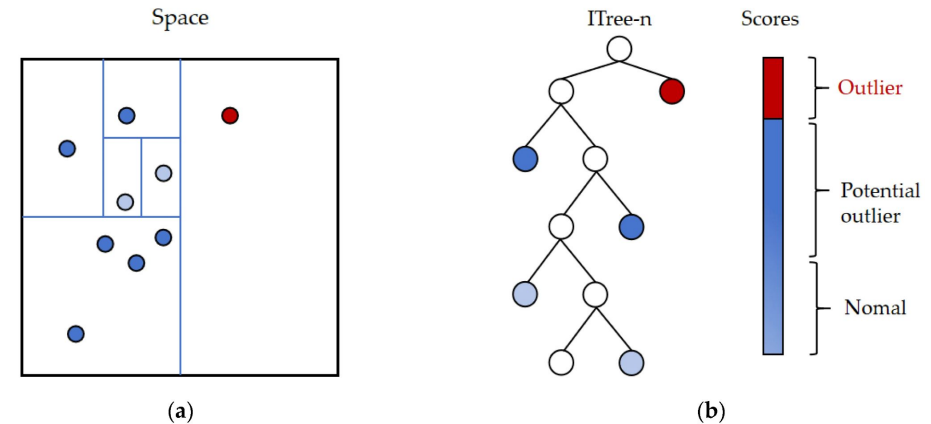
(a) Isolating x_i



(b) Isolating x_o

Background - Isolation Forest

- Training stage

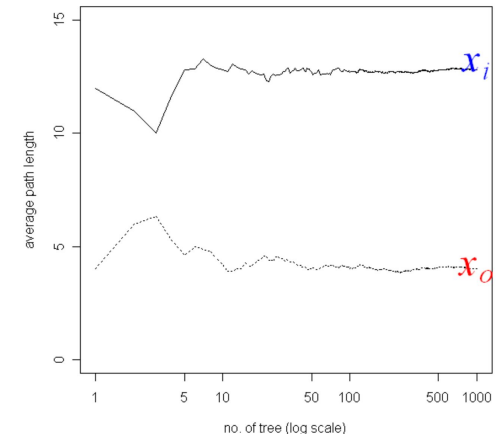
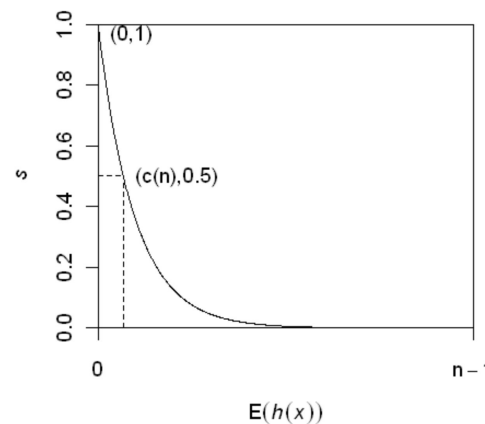


- Evaluating stage

- **Anomaly score**

- Average path length

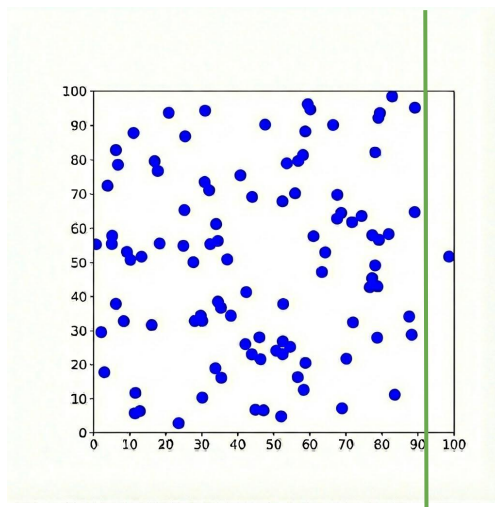
$$s(\mathbf{x}, n) = 2^{-\frac{E(h(\mathbf{x}))}{c(n)}}$$



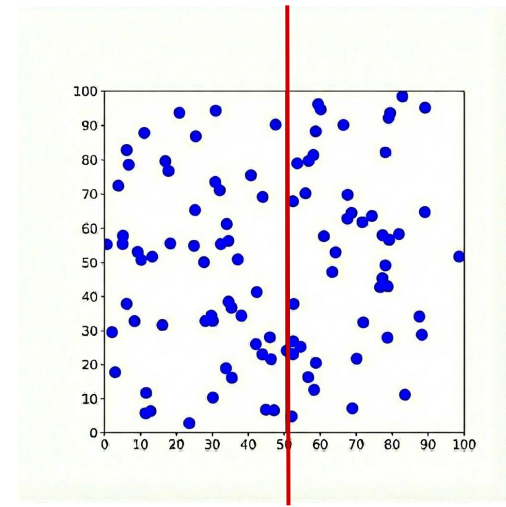
Background - DIFFI

- **Depth-based Isolation Forest Feature Importance (DIFFI)**
- *Model-specific post-hoc* explainability method for Isolation Forest
- Assumptions
 - the importance of a feature is based on at which depth isolate an outlier/inlier
 - a good split produces high imbalance on anomalous point

Good split



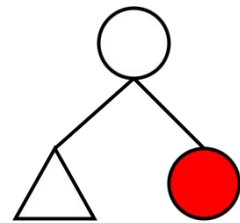
Bad split



Research objective

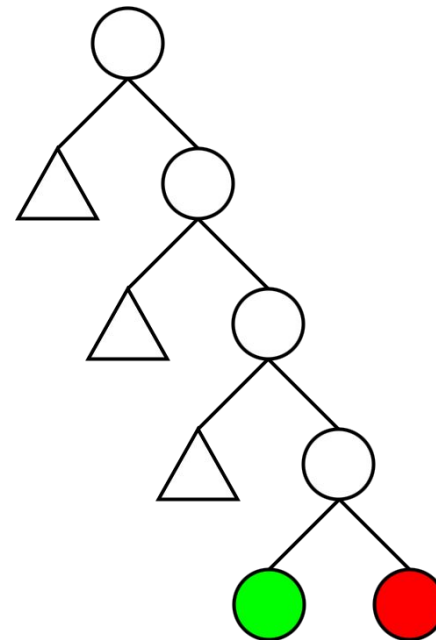


- Propose a technique to prune Isolation Forest
- Motivations:
 - **Presence of the non-informative trees**
 - Unsupervised setting of the Anomaly Detection task



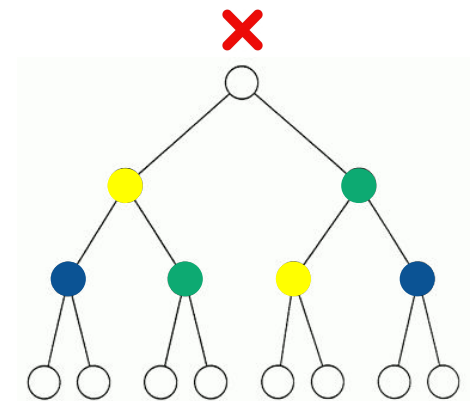
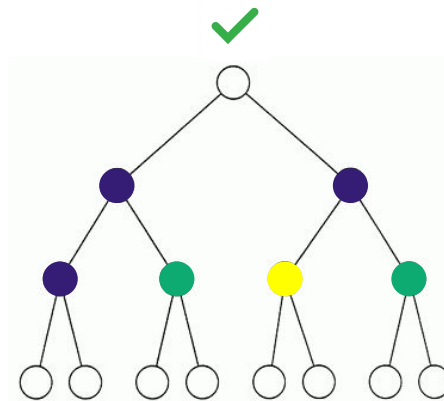
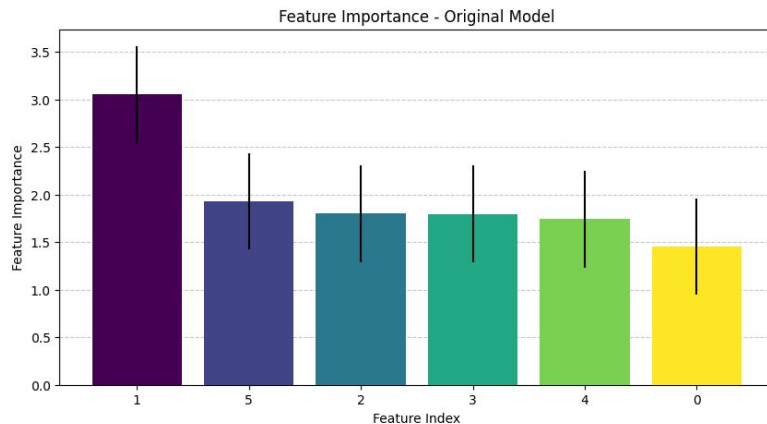
Legend:

- Outliers
- Inliers



DIFFI-based pruning

- How to identify non-informative tree?



Feature Selection

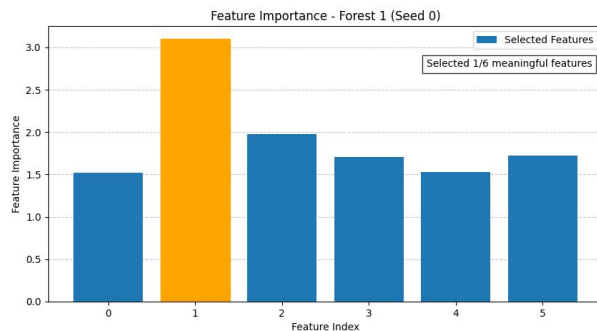
Usage Score

DIFFI-based pruning

Feature Selection

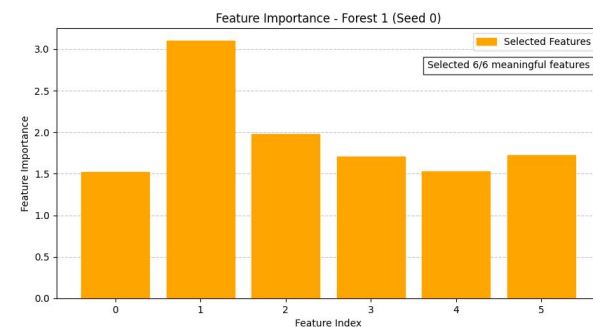
Elbow approach

1. Sorting them in decreasing order
2. Computing gaps
3. Identify the maximum gap
4. Selecting the features before such gap



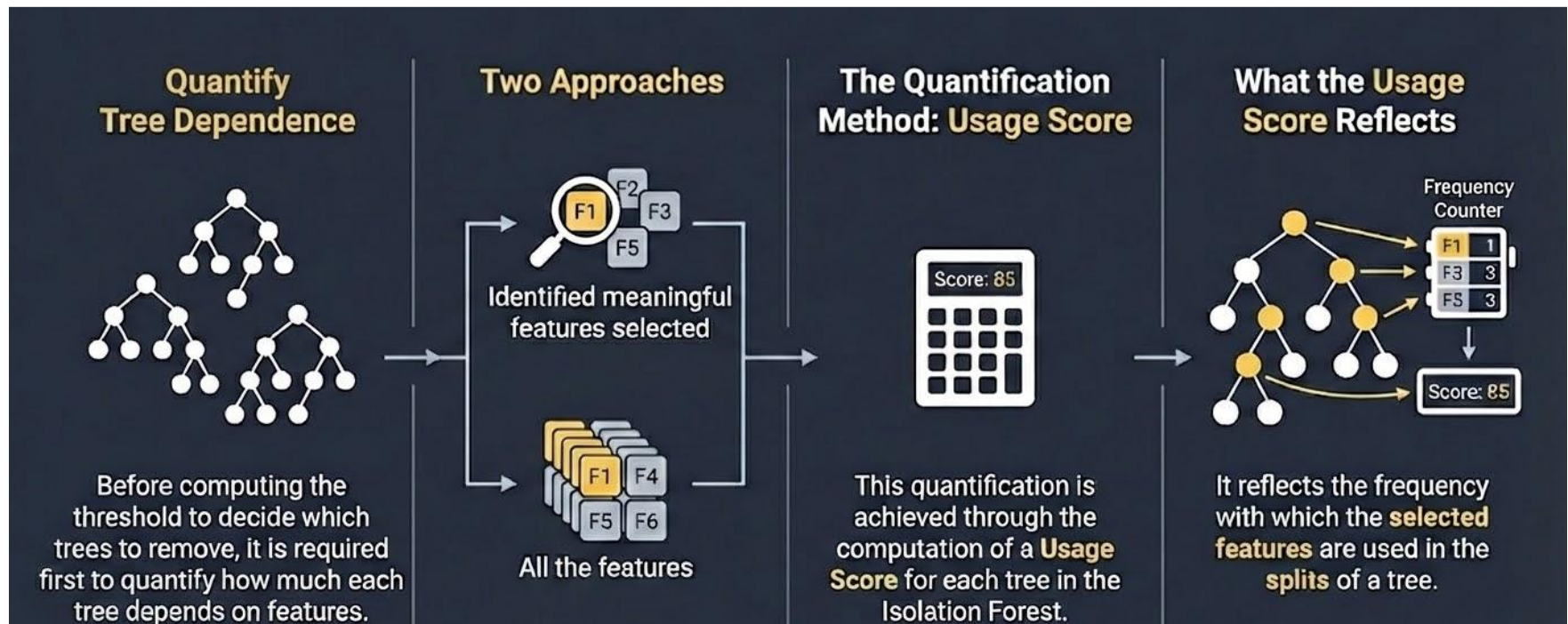
Soft approach

1. Including all features by weighting their contributions with the corresponding GFI values



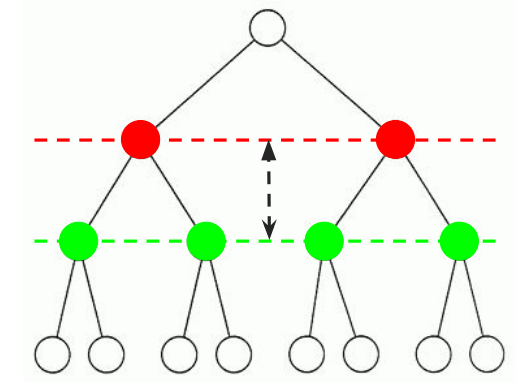
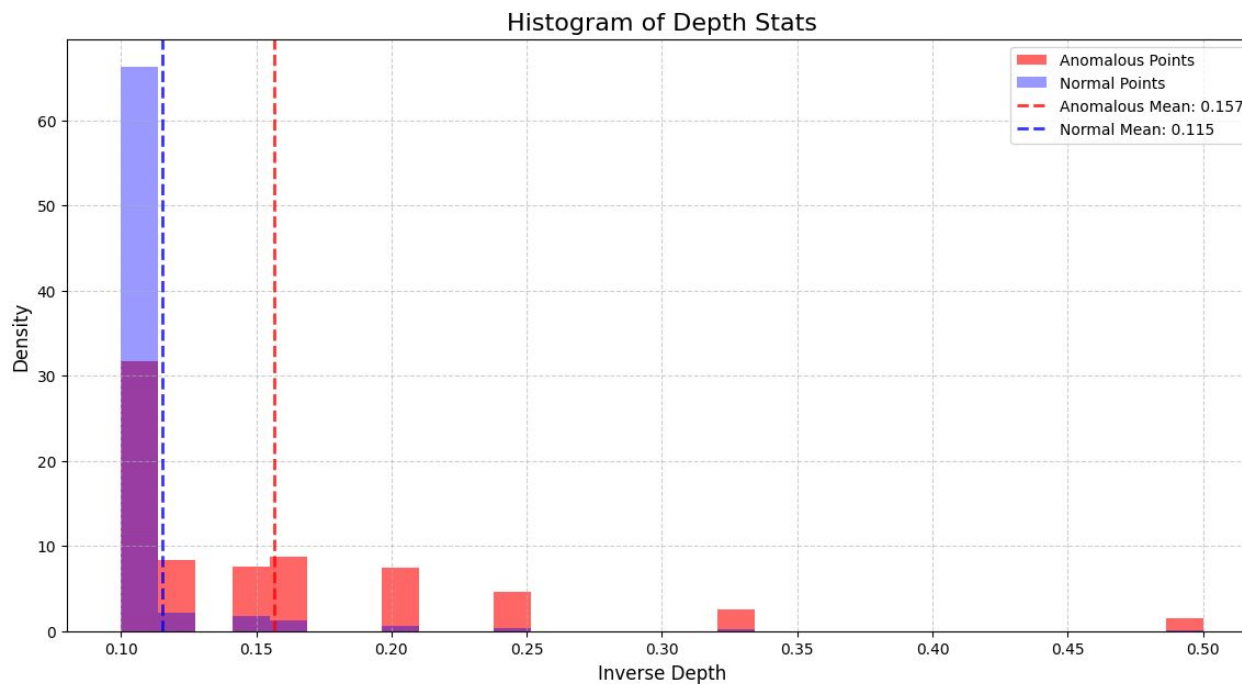
DIFFI-based pruning

Usage Score



Depth-based pruning

- How to identify non-informative tree?

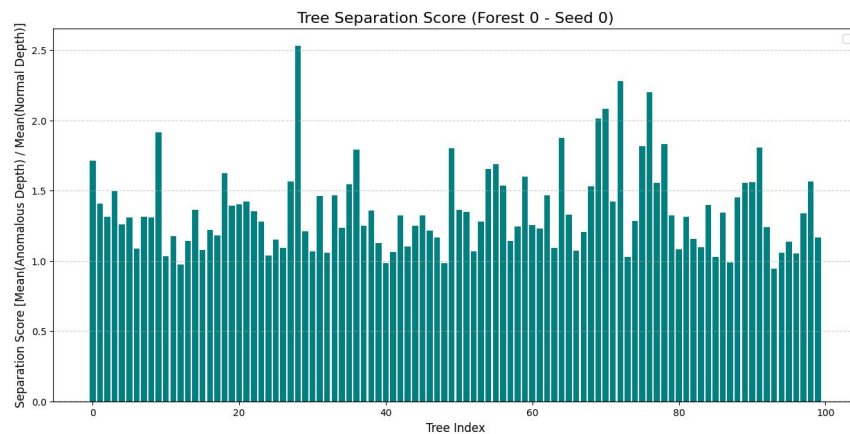


Depth-based pruning

Depth scores

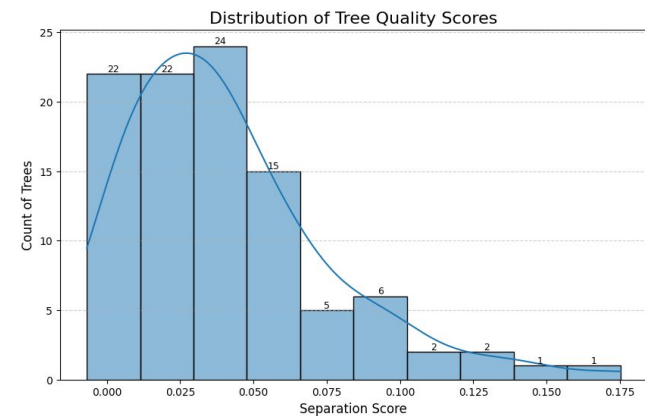
Separation Ratio

$$SR_i = \frac{D_{outliers}^i}{D_{inliers}^i}$$

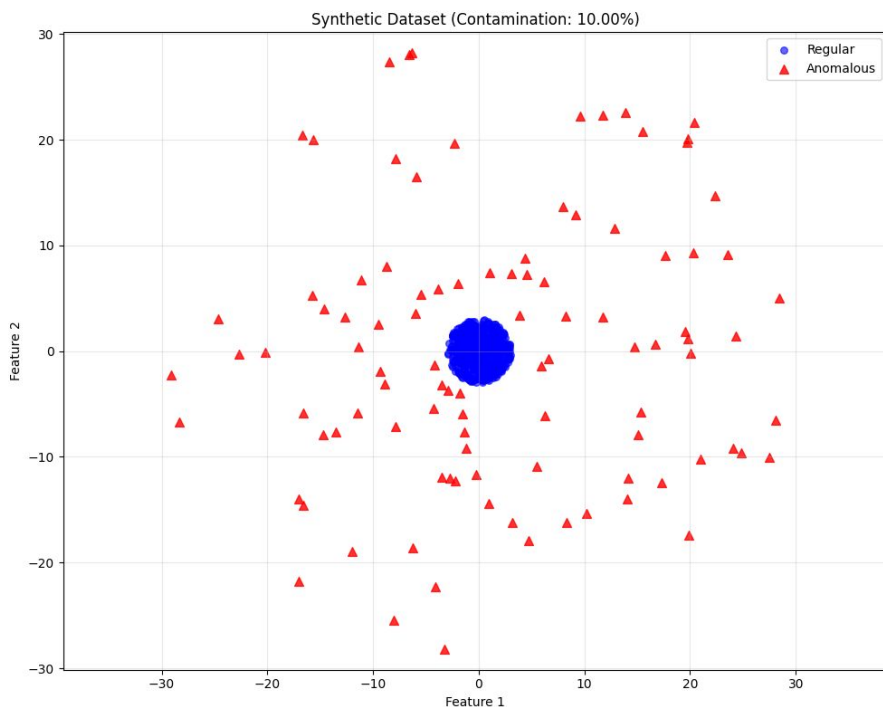


Anomaly Depth

$$D_{outliers}^i$$



Synthetic



- 2 meaningful features
- 4 irrelevant features



Outliers Detection DataSets (ODDS)

Dataset	#points	#dim.	#outliers (%)
Vowels	1456	12	50 (3.4%)
Cardio	1831	21	176 (9.6%)
Thyroid	3772	6	93 (2.5%)
Satimage-2	5803	36	71 (1.2%)
Letter Recognition	1600	32	100 (6.25%)
Pima	768	8	268 (35%)
Satellite	6435	36	2036 (32%)
BreastW	683	9	239 (35%)
Ionosphere	351	33	126 (36%)
Mnist	7603	100	700 (9.2%)
Optdigits	5216	64	150 (3%)
Anthyroid	7200	6	534 (7.42%)
Pendigits	6870	16	156 (2.27%)
Wine	129	13	10 (7.7%)
Vertebral	240	6	30 (12.5%)

Results - DIFFI-based

Elbow approach

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.5289	+21.59%	0.5218	+19.95%	0.5371	+23.47%
annthyroid	0.3051	0.3625	+18.8%	0.3603	+18.1%	0.3929	+28.8%
breastw	0.9714	0.9698	-0.2%	0.9702	-0.1%	0.9651	-0.6%
cardio	0.5685	0.5613	-1.3%	0.5644	-0.7%	0.5302	-6.7%
ionosphere	0.8235	0.8223	-0.1%	0.8226	-0.1%	0.8185	-0.6%
letter	0.0988	0.0992	+0.4%	0.0997	+0.9%	0.0991	+0.3%
mnist	0.2777	0.2779	+0.1%	0.2778	+0.0%	0.2702	-2.7%
optdigits	0.0540	0.0500	-7.4%	0.0503	-6.8%	0.0461	-14.6%
pendigits	0.2795	0.2619	-6.3%	0.2639	-5.6%	0.2407	-13.9%
pima	0.5055	0.4985	-1.4%	0.4992	-1.2%	0.4891	-3.2%
satellite	0.6729	0.6687	-0.6%	0.6692	-0.5%	0.6521	-3.1%
satimage-2	0.9089	0.8981	-1.2%	0.8994	-1.0%	0.8709	-4.2%
thyroid	0.5763	0.6443	+11.8%	0.6427	+11.5%	0.6640	+15.2%
vertebral	0.1129	0.1134	+0.4%	0.1132	+0.3%	0.1151	+1.9%
vowels	0.1712	0.1628	-4.9%	0.1629	-4.8%	0.1415	-17.3%
wine	0.3228	0.3154	-2.3%	0.3148	-2.5%	0.2891	-10.4%

Soft approach

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.5417	+24.53%	0.5373	+23.52%	0.5493	+26.28%
annthyroid	0.3051	0.3559	+16.6%	0.3535	+15.9%	0.3847	+26.1%
breastw	0.9714	0.9700	-0.1%	0.9704	-0.1%	0.9652	-0.6%
cardio	0.5685	0.5714	+0.5%	0.5726	+0.7%	0.5468	-3.8%
ionosphere	0.8235	0.8232	-0.0%	0.8231	-0.0%	0.8204	-0.4%
letter	0.0988	0.0988	+0.0%	0.0988	+0.0%	0.0977	-1.1%
mnist	0.2777	0.2850	+2.6%	0.2844	+2.4%	0.2807	+1.1%
optdigits	0.0540	0.0487	-9.8%	0.0493	-8.7%	0.0452	-16.3%
pendigits	0.2795	0.2658	-4.9%	0.2653	-5.1%	0.2413	-13.7%
pima	0.5055	0.4991	-1.3%	0.5004	-1.0%	0.4892	-3.2%
satellite	0.6729	0.6704	-0.4%	0.6704	-0.4%	0.6564	-2.4%
satimage-2	0.9089	0.9030	-0.6%	0.9033	-0.6%	0.8762	-3.6%
thyroid	0.5763	0.6423	+11.4%	0.6387	+10.8%	0.6588	+14.3%
vertebral	0.1129	0.1134	+0.4%	0.1125	-0.4%	0.1169	+3.5%
vowels	0.1712	0.1641	-4.2%	0.1635	-4.5%	0.1472	-14.0%
wine	0.3228	0.3092	-4.2%	0.3098	-4.0%	0.2876	-10.9%

Results - DIFFI-based

Elbow approach

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.5289	+21.59%	0.5218	+19.95%	0.5371	+23.47%
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cardio	0.5685	0.5613	-1.3%	0.5644	-0.7%	0.5302	-6.7%
ionosphere	0.8235	0.8223	-0.1%	0.8226	-0.1%	0.8185	-0.6%
letter	0.0988	0.0992	+0.4%	0.0997	+0.9%	0.0991	+0.3%
mnist	0.2777	0.2779	+0.1%	0.2778	+0.0%	0.2702	-2.7%
optdigits	0.0540	0.0500	-7.4%	0.0503	-6.8%	0.0461	-14.6%
pendigits	0.2795	0.2619	-6.3%	0.2639	-5.6%	0.2407	-13.9%
pima	0.5055	0.4985	-1.4%	0.4992	-1.2%	0.4891	-3.2%
satellite	0.6729	0.6687	-0.6%	0.6692	-0.5%	0.6521	-3.1%
satimage-2	0.9089	0.8981	-1.2%	0.8994	-1.0%	0.8709	-4.2%
thyroid	0.5763	0.6443	+11.8%	0.6427	+11.5%	0.6640	+15.2%
vertebral	0.1129	0.1134	+0.4%	0.1132	+0.3%	0.1151	+1.9%
vowels	0.1712	0.1628	-4.9%	0.1629	-4.8%	0.1415	-17.3%
wine	0.3228	0.3154	-2.3%	0.3148	-2.5%	0.2891	-10.4%

Soft approach

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.5417	+24.53%	0.5373	+23.52%	0.5493	+26.28%
annthyroid	0.3051	0.3559	+16.6%	0.3535	+15.9%	0.3847	+26.1%
breastw	0.9714	0.9700	-0.1%	0.9704	-0.1%	0.9652	-0.6%
cardio	0.5685	0.5714	+0.5%	0.5726	+0.7%	0.5468	-3.8%
ionosphere	0.8235	0.8232	-0.0%	0.8231	-0.0%	0.8204	-0.4%
letter	0.0988	0.0988	+0.0%	0.0988	+0.0%	0.0977	-1.1%
mnist	0.2777	0.2850	+2.6%	0.2844	+2.4%	0.2807	+1.1%
optdigits	0.0540	0.0487	-9.8%	0.0493	-8.7%	0.0452	-16.3%
pendigits	0.2795	0.2658	-4.9%	0.2653	-5.1%	0.2413	-13.7%
pima	0.5055	0.4991	-1.3%	0.5004	-1.0%	0.4892	-3.2%
satellite	0.6729	0.6704	-0.4%	0.6704	-0.4%	0.6564	-2.4%
satimage-2	0.9089	0.9030	-0.6%	0.9033	-0.6%	0.8762	-3.6%
thyroid	0.5763	0.6423	+11.4%	0.6387	+10.8%	0.6588	+14.3%
vertebral	0.1129	0.1134	+0.4%	0.1125	-0.4%	0.1169	+3.5%
vowels	0.1712	0.1641	-4.2%	0.1635	-4.5%	0.1472	-14.0%
wine	0.3228	0.3092	-4.2%	0.3098	-4.0%	0.2876	-10.9%

Results - DIFFI-based

Elbow approach

(b) Average Precision

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Soft approach

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synthetic	0.4350	0.5417	+24.53%	0.5373	+23.52%	0.5493	+26.28%
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Results - DIFFI-based

Elbow approach

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Soft approach

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letter	0.0988	0.0988	+0.0%	0.0988	+0.0%	0.0977	-1.1%
mnist	0.2777	0.2850	+2.6%	0.2844	+2.4%	0.2807	+1.1%
optdigits	0.0540	0.0487	-9.8%	0.0493	-8.7%	0.0452	-16.3%
pendigits	0.2795	0.2658	-4.9%	0.2653	-5.1%	0.2413	-13.7%
pima	0.5055	0.4991	-1.3%	0.5004	-1.0%	0.4892	-3.2%
satellite	0.6729	0.6704	-0.4%	0.6704	-0.4%	0.6564	-2.4%
satimage-2	0.9089	0.9030	-0.6%	0.9033	-0.6%	0.8762	-3.6%
thyroid	0.5763	0.6423	+11.4%	0.6387	+10.8%	0.6588	+14.3%
vertebral	0.1129	0.1134	+0.4%	0.1125	-0.4%	0.1169	+3.5%
vowels	0.1712	0.1641	-4.2%	0.1635	-4.5%	0.1472	-14.0%
wine	0.3228	0.3092	-4.2%	0.3098	-4.0%	0.2876	-10.9%

Results - Depth-based

Separation ratio

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.7013	+61.2%	0.6364	+46.3%	0.7338	+68.7%
annthyroid	0.3051	0.4437	+45.4%	0.4267	+39.9%	0.4833	+58.4%
breastw	0.9714	0.9795	+0.8%	0.9797	+0.8%	0.9779	+0.7%
cardio	0.5685	0.7166	+26.0%	0.7009	+23.3%	0.7539	+32.6%
ionosphere	0.8235	0.8401	+2.0%	0.8406	+2.1%	0.8466	+2.8%
letter	0.0988	0.1261	+27.6%	0.1208	+22.3%	0.1439	+45.6%
mnist	0.2777	0.4720	+70.0%	0.4435	+59.7%	0.5239	+88.7%
optdigits	0.0540	0.2665	+393.5%	0.1787	+230.9%	0.3710	+587.0%
pendigits	0.2795	0.5108	+82.8%	0.4839	+73.1%	0.5428	+94.2%
pima	0.5055	0.5407	+7.0%	0.5405	+6.9%	0.5523	+9.3%
satellite	0.6729	0.7205	+7.1%	0.7191	+6.9%	0.7216	+7.2%
satimage-2	0.9089	0.9252	+1.8%	0.9253	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8035	+39.4%	0.7888	+36.9%	0.8195	+42.2%
vertebral	0.1129	0.1503	+33.1%	0.1399	+23.9%	0.1678	+48.6%
vowels	0.1712	0.3531	+106.2%	0.3644	+112.8%	0.3324	+94.2%
wine	0.3228	0.6345	+96.6%	0.5854	+81.3%	0.6834	+111.7%

Anomaly depth

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.6912	+58.9%	0.6235	+43.3%	0.7287	+67.5%
annthyroid	0.3051	0.4419	+44.8%	0.4251	+39.3%	0.4812	+57.7%
breastw	0.9714	0.9787	+0.8%	0.9787	+0.8%	0.9769	+0.6%
cardio	0.5685	0.7130	+25.4%	0.6965	+22.5%	0.7488	+31.7%
ionosphere	0.8235	0.8382	+1.8%	0.8382	+1.8%	0.8439	+2.5%
letter	0.0988	0.1220	+23.5%	0.1174	+18.8%	0.1396	+41.3%
mnist	0.2777	0.4672	+68.2%	0.4351	+56.7%	0.5175	+86.3%
optdigits	0.0540	0.2547	+371.7%	0.1689	+212.8%	0.3544	+556.3%
pendigits	0.2795	0.5123	+83.3%	0.4861	+73.9%	0.5475	+95.9%
pima	0.5055	0.5361	+6.1%	0.5346	+5.8%	0.5430	+7.4%
satellite	0.6729	0.7172	+6.6%	0.7157	+6.4%	0.7199	+7.0%
satimage-2	0.9089	0.9253	+1.8%	0.9250	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8036	+39.4%	0.7888	+36.9%	0.8190	+42.1%
vertebral	0.1129	0.1486	+31.6%	0.1393	+23.4%	0.1649	+46.1%
vowels	0.1712	0.3495	+104.1%	0.3239	+89.2%	0.3644	+112.8%
wine	0.3228	0.6256	+93.8%	0.5856	+81.4%	0.6865	+112.7%

Results - Depth-based

Separation ratio

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.7013	+61.2%	0.6364	+46.3%	0.7338	+68.7%
annthyroid	0.3051	0.4437	+45.4%	0.4267	+39.9%	0.4833	+58.4%
breastw	0.9714	0.9795	+0.8%	0.9797	+0.8%	0.9779	+0.7%
cardio	0.5685	0.7166	+26.0%	0.7009	+23.3%	0.7539	+32.6%
ionosphere	0.8235	0.8401	+2.0%	0.8406	+2.1%	0.8466	+2.8%
letter	0.0988	0.1261	+27.6%	0.1208	+22.3%	0.1439	+45.6%
mnist	0.2777	0.4720	+70.0%	0.4435	+59.7%	0.5239	+88.7%
optdigits	0.0540	0.2665	+393.5%	0.1787	+230.9%	0.3710	+587.0%
pendigits	0.2795	0.5108	+82.8%	0.4839	+73.1%	0.5428	+94.2%
pima	0.5055	0.5407	+7.0%	0.5405	+6.9%	0.5523	+9.3%
satellite	0.6729	0.7205	+7.1%	0.7191	+6.9%	0.7216	+7.2%
satimage-2	0.9089	0.9252	+1.8%	0.9253	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8035	+39.4%	0.7888	+36.9%	0.8195	+42.2%
vertebral	0.1129	0.1503	+33.1%	0.1399	+23.9%	0.1678	+48.6%
vowels	0.1712	0.3531	+106.2%	0.3644	+112.8%	0.3324	+94.2%
wine	0.3228	0.6345	+96.6%	0.5854	+81.3%	0.6834	+111.7%

Anomaly depth

(b) Average Precision

Dataset	Original	Mean		P50		P80	
		Pruned	Var.	Pruned	Var.	Pruned	Var.
synthetic	0.4350	0.6912	+58.9%	0.6235	+43.3%	0.7287	+67.5%
annthyroid	0.3051	0.4419	+44.8%	0.4251	+39.3%	0.4812	+57.7%
breastw	0.9714	0.9787	+0.8%	0.9787	+0.8%	0.9769	+0.6%
cardio	0.5685	0.7130	+25.4%	0.6965	+22.5%	0.7488	+31.7%
ionosphere	0.8235	0.8382	+1.8%	0.8382	+1.8%	0.8439	+2.5%
letter	0.0988	0.1220	+23.5%	0.1174	+18.8%	0.1396	+41.3%
mnist	0.2777	0.4672	+68.2%	0.4351	+56.7%	0.5175	+86.3%
optdigits	0.0540	0.2547	+371.7%	0.1689	+212.8%	0.3544	+556.3%
pendigits	0.2795	0.5123	+83.3%	0.4861	+73.9%	0.5475	+95.9%
pima	0.5055	0.5361	+6.1%	0.5346	+5.8%	0.5430	+7.4%
satellite	0.6729	0.7172	+6.6%	0.7157	+6.4%	0.7199	+7.0%
satimage-2	0.9089	0.9253	+1.8%	0.9250	+1.8%	0.9223	+1.5%
thyroid	0.5763	0.8036	+39.4%	0.7888	+36.9%	0.8190	+42.1%
vertebral	0.1129	0.1486	+31.6%	0.1393	+23.4%	0.1649	+46.1%
vowels	0.1712	0.3495	+104.1%	0.3239	+89.2%	0.3644	+112.8%
wine	0.3228	0.6256	+93.8%	0.5856	+81.4%	0.6865	+112.7%

Results comparison

(b) Average Precision

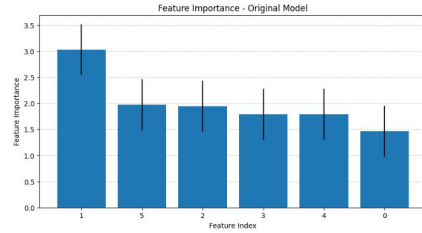
Dataset	Original	Random	DIFFI-based		Depth-based	
			Elbow	Soft	Separation Ratio	Anomaly Depth
synthetic	0.4350	0.4048	0.5435	0.5541	0.7338	0.7287
annthyroid	0.3051	0.3029	0.3929	0.3847	0.4833	0.4812
breastw	0.9714	0.9690	0.9714	0.9714	0.9797	0.9787
cardio	0.5685	0.5550	0.5685	0.5726	0.7539	0.7488
ionosphere	0.8235	0.8218	0.8235	0.8235	0.8466	0.8439
letter	0.0988	0.0987	0.0997	0.0988	0.1439	0.1396
mnist	0.2777	0.2724	0.2779	0.2850	0.5239	0.5175
optdigits	0.0540	0.0539	0.0540	0.0540	0.3710	0.3544
pendigits	0.2795	0.2716	0.2795	0.2795	0.5428	0.5475
pima	0.5055	0.5030	0.5055	0.5055	0.5523	0.5430
satellite	0.6729	0.6689	0.6729	0.6729	0.7216	0.7199
satimage-2	0.9089	0.9013	0.9089	0.9089	0.9253	0.9253
thyroid	0.5763	0.5685	0.6640	0.6588	0.8195	0.8190
vertebral	0.1129	0.1170	0.1151	0.1169	0.1678	0.1649
vowels	0.1712	0.1596	0.1712	0.1712	0.3644	0.3644
wine	0.3228	0.3191	0.3228	0.3228	0.6834	0.6865

Results comparison - Distributions

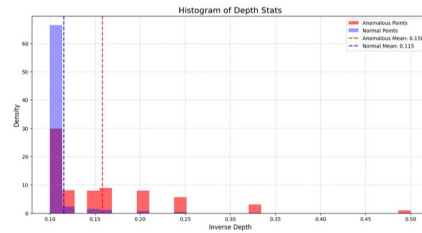


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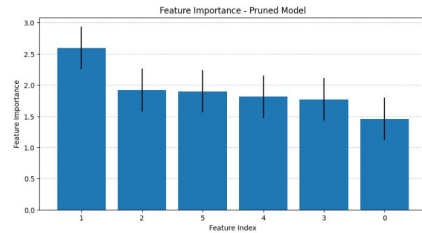
Thyroid



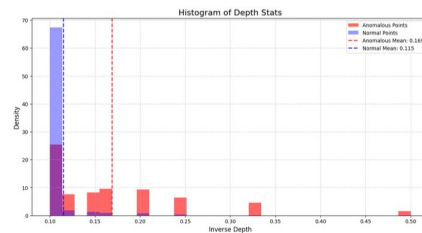
(a) GFI Distribution for Original Model.



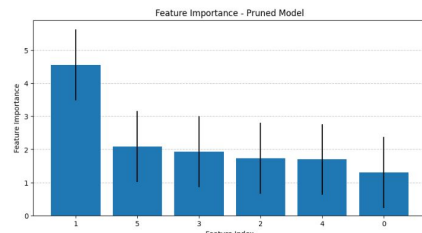
(b) Depth Distribution for Original Model.



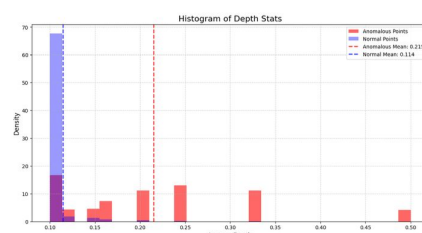
(c) GFI Distribution for DIFFI-based Pruned Model.



(d) Depth Distribution for DIFFI-based Pruned Model.

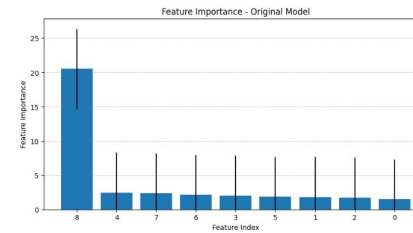


(e) GFI Distribution for Depth-based Pruned Model.

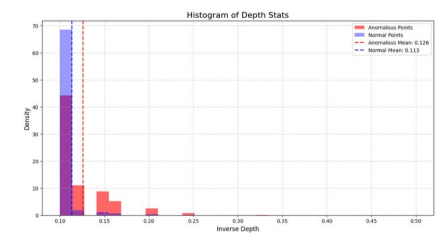


(f) Depth Distribution for Depth-based Pruned Model.

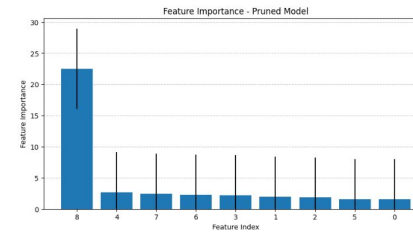
Breastw



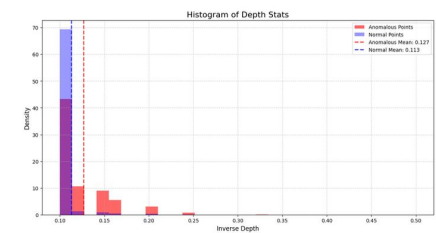
(a) GFI Distribution for Original Model.



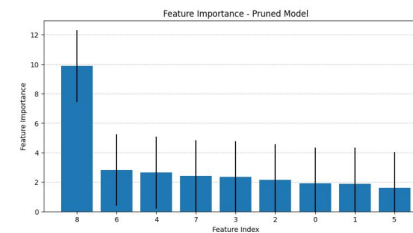
(b) Depth Distribution for Original Model.



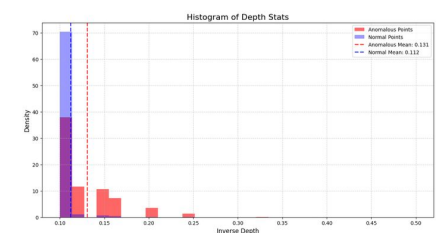
(c) GFI Distribution for DIFFI-based Pruned Model.



(d) Depth Distribution for DIFFI-based Pruned Model.



(e) GFI Distribution for Depth-based Pruned Model.



(f) Depth Distribution for Depth-based Pruned Model.

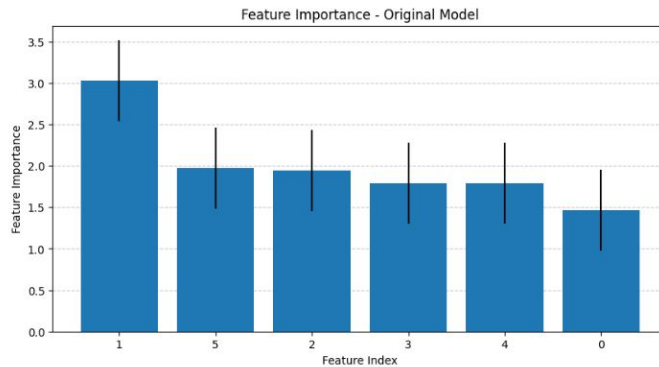


Results comparison - Distributions

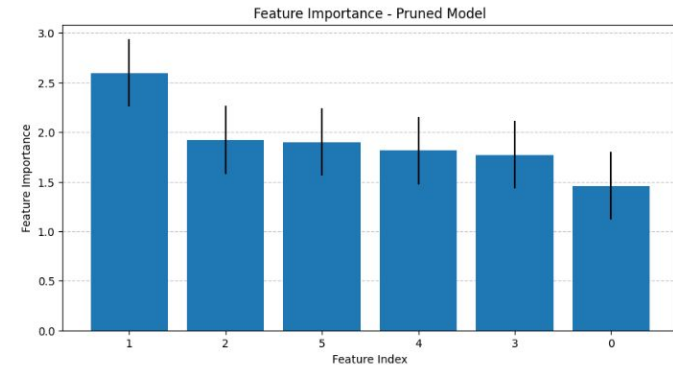


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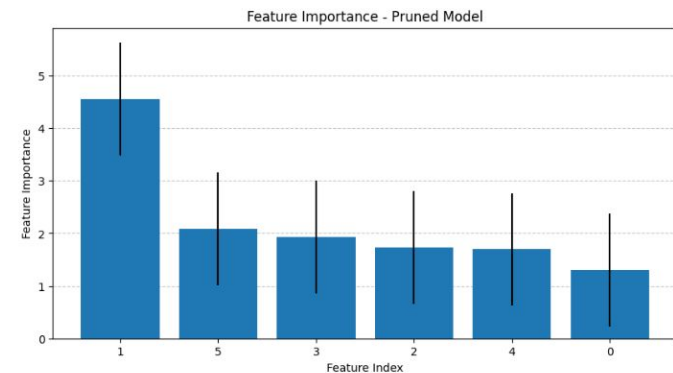
Thyroid



(a) GFI Distribution for Original Model.



(c) GFI Distribution for DIFFI-based Pruned Model.



(e) GFI Distribution for Depth-based Pruned Model.

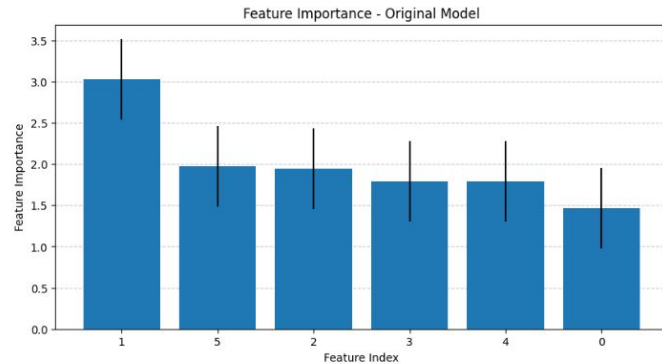


Results comparison - Distributions

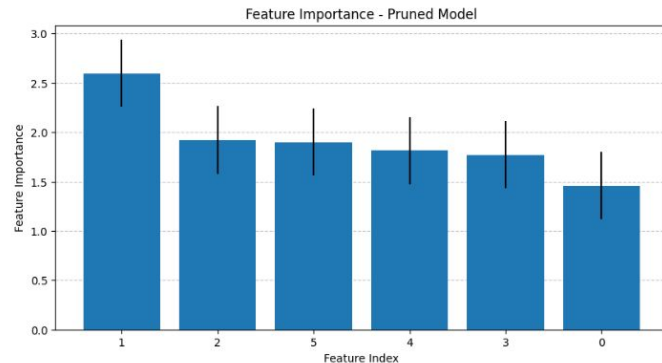


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Thyroid

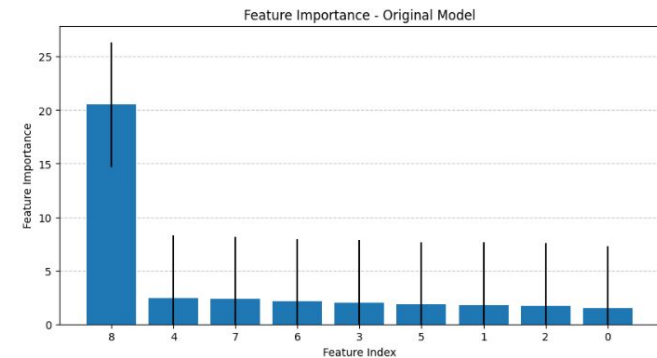


(a) GFI Distribution for Original Model.

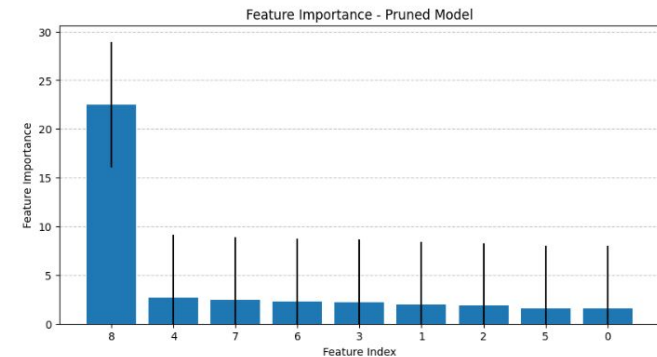


(c) GFI Distribution for DIFFI-based Pruned Model.

Breastw



(a) GFI Distribution for Original Model.

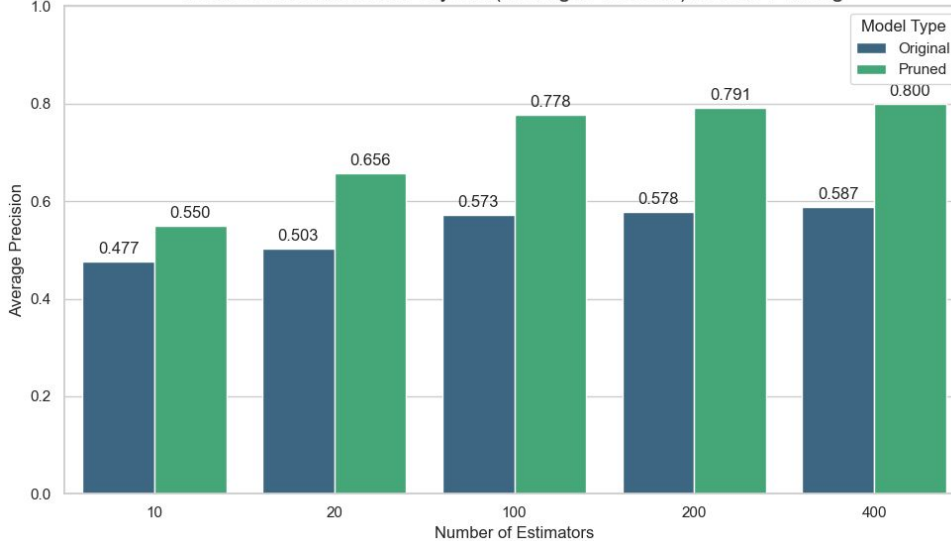


(c) GFI Distribution for DIFFI-based Pruned Model.

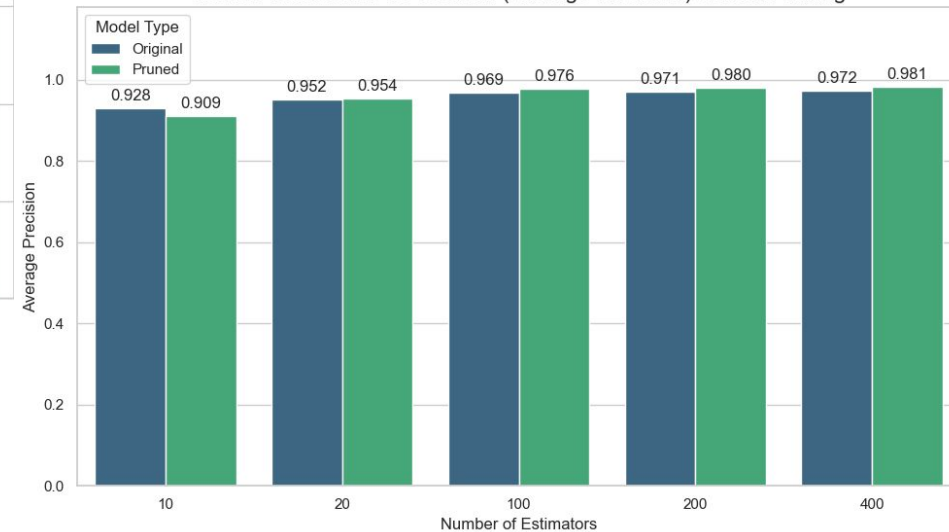


Impact of the number of estimators

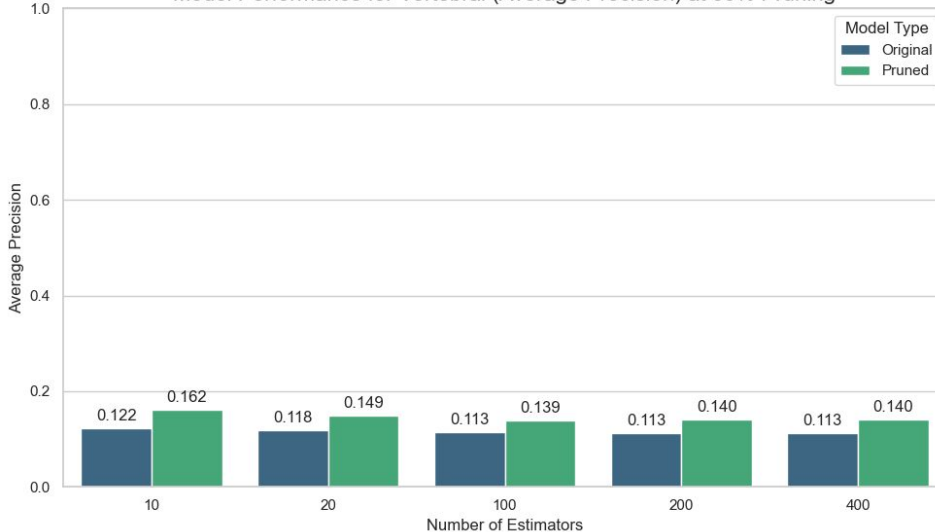
Model Performance for Thyroid (Average Precision) at 50% Pruning



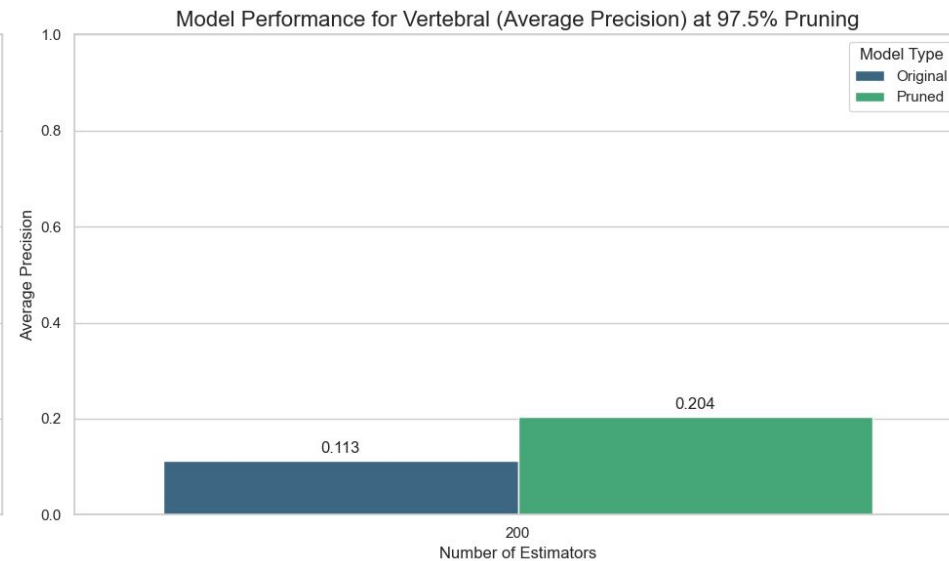
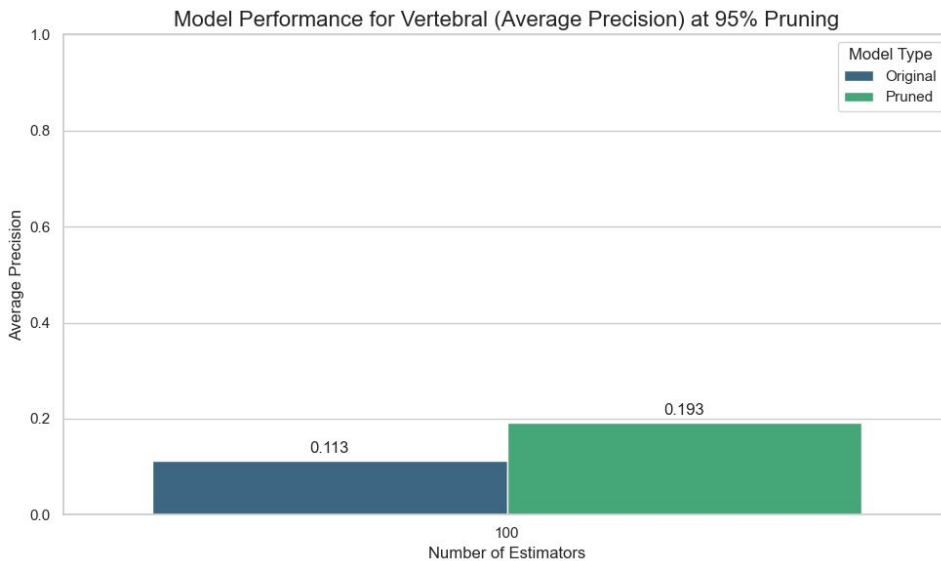
Model Performance for Breastw (Average Precision) at 50% Pruning



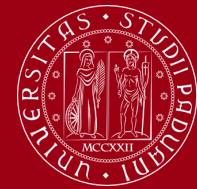
Model Performance for Vertebral (Average Precision) at 50% Pruning



Impact of the number of estimators



Conclusion



01

Effective pruning strategy

02

Reduction in ensemble size

03

Impact of the number of
estimators

04

Complex relationship with
DIFFI



Future works



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01

Perform experiments on larger
number of dataset

02

Determine the computational
complexity empirically

03

Deeper exploration of DIFFI

04

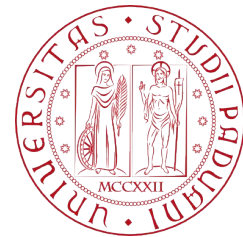
Expand the experiments about
the impact of the number of
estimators

05

Analysis and identification of
the redundant trees



Thank you



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Background - Isolation Forest

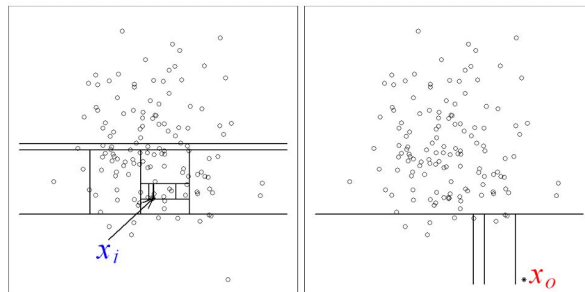
- Training stage
 - Complexity: $O(t\psi \log \psi)$

Algorithm 1 : $iForest(X, t, \psi)$

Inputs: X - input data, t - number of trees, ψ - sub-sampling size

Output: a set of t $iTrees$

- 1: **Initialize** $Forest$
 - 2: set height limit $l = ceiling(\log_2 \psi)$
 - 3: **for** $i = 1$ to t **do**
 - 4: $X' \leftarrow sample(X, \psi)$
 - 5: $Forest \leftarrow Forest \cup iTree(X', 0, l)$
 - 6: **end for**
 - 7: **return** $Forest$
-



(a) Isolating x_i

(b) Isolating x_o

Algorithm 2 : $iTree(X, e, l)$

Inputs: X - input data, e - current tree height, l - height limit

Output: an $iTree$

- 1: **if** $e \geq l$ or $|X| \leq 1$ **then**
 - 2: return $exNode\{Size \leftarrow |X|\}$
 - 3: **else**
 - 4: let Q be a list of attributes in X
 - 5: randomly select an attribute $q \in Q$
 - 6: randomly select a split point p from max and min values of attribute q in X
 - 7: $X_l \leftarrow filter(X, q < p)$
 - 8: $X_r \leftarrow filter(X, q \geq p)$
 - 9: return $inNode\{Left \leftarrow iTree(X_l, e + 1, l),$
 - 10: $Right \leftarrow iTree(X_r, e + 1, l),$
 - 11: $SplitAtt \leftarrow q,$
 - 12: $SplitValue \leftarrow p\}$
 - 13: **end if**
-

Liu, F. T., Ting, K., & Zhou, Z.-H. (2009). Isolation forest

Background - Isolation Forest

- Evaluating stage

- **Anomaly score**

- Average path length
- Normalization constant

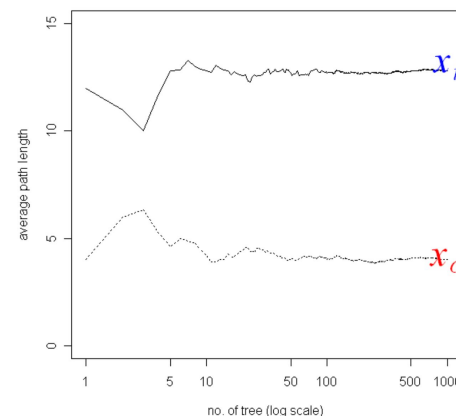
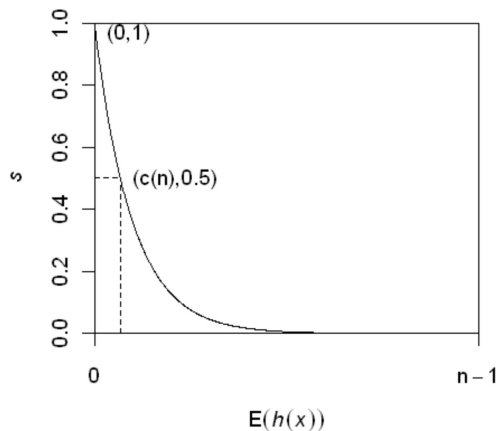
$$s(\mathbf{x}, n) = 2^{-\frac{E(h(\mathbf{x}))}{c(n)}}$$

$$c(n) = 2H(n-1) - \left(\frac{2(n-1)}{n}\right)$$

- Harmonic number

$$H(n) = \sum_{i=1}^n \frac{1}{i}$$

- Complexity: **$O(Nt \log \psi)$**

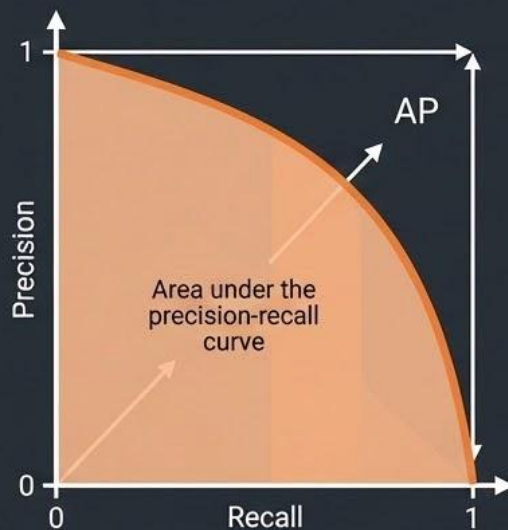


Liu, F. T., Ting, K., & Zhou, Z.-H. (2009). Isolation forest

Average Precision

CORE CONCEPT

A single score summarizing model performance by calculating the area under its **precision-recall** curve (Buitinck et al., 2013).



CALCULATION PROCESS

1. **RANK** instances by anomaly scores (descending).
2. **CALCULATE** precision at various threshold levels.
3. **AVERAGE** precision values, weighting for greater recall increases.

Threshold-independent

UTILITY & ADVANTAGES

-  **COMPLEMENT** to F1-score, avoiding threshold sensitivity issues.
-  Provides a **COMPREHENSIVE EVALUATION** of anomaly detection effectiveness.
-  Valuable for assessing **ROBUSTNESS** and **RELIABILITY** in different operating conditions.

AVERAGE PRECISION IN CONTEXT

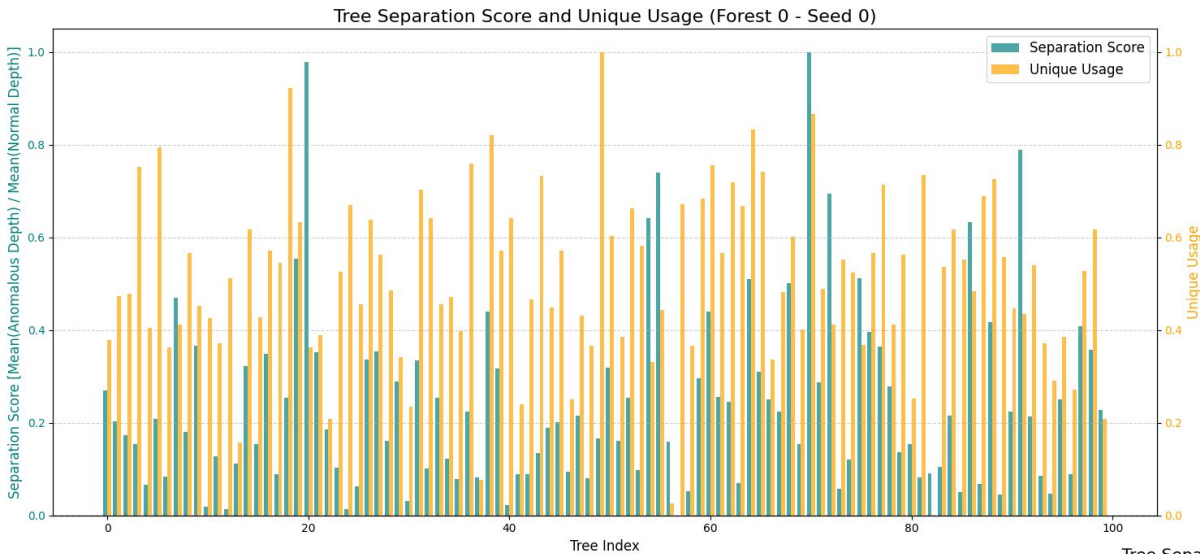
METRIC	DESCRIPTION	KEY STRENGTH
Average Precision (AP)	Area under Precision-Recall Curve, threshold-independent.	Comprehensive, robust evaluation across all thresholds.
F1-Score	Harmonic mean of Precision and Recall at a specific threshold.	Good for imbalanced data at a single point.
Precision	Fraction of true positives among all positive predictions.	Minimizing false positives.
Recall	Fraction of true positives among all actual positives.	Minimizing false negatives (missed anomalies).

Source: Buitinck et al., 2013. Anomaly Detection Evaluation.

Results comparison - Tree scores



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Thyroid

Breastw

